Nonhomogeneous Equations & Undetermined Coefficients

ANSWER KEY

Complementary function and particular solution

1 For $y'' - 3y' + 2y = 4e^{3x}$, find the complementary function y_c.

$r^2 - 3r + 2 = (r-1)(r-2) = 0 \Rightarrow r = 1, 2$. So $y_c = C_1 e^x + C_2 e^{2x}$.

2 Explain why the general solution of a nonhomogeneous linear equation is $y = y_c + y_p$.

If y_p is any particular solution and y_c solves the associated homogeneous equation, then $y_c + y_p$ solves the nonhomogeneous equation (by linearity/superposition) and carries enough arbitrary constants to match any initial condition — so it is the general solution.

3 For $y'' + 4y = 8$, find y_c and a constant particular solution y_p.

$r^2 + 4 = 0 \Rightarrow r = \pm 2i$, so $y_c = C_1 \cos 2x + C_2 \sin 2x$. Try $y_p = A$: $y_p'' = 0$, so $4A = 8 \Rightarrow A = 2$. Thus $y_p = 2$.

Choosing a trial form (no resonance)

4 Find a particular solution of $y'' - 3y' + 2y = 4x$.

Try $y_p = Ax + B$: $y_p' = A$, $y_p'' = 0$. Substituting: $-3A + 2(Ax + B) = 4x \Rightarrow 2Ax + (2B - 3A) = 4x$. So $2A = 4 \Rightarrow A = 2$, and $2B - 3A = 0 \Rightarrow B = 3$. Thus $y_p = 2x + 3$.

5 Find a particular solution of $y'' + y' - 2y = 6e^{3x}$ (note $r = 3$ is not a root of $r^2 + r - 2 = 0$).

Try $y_p = Ae^{3x}$: $y_p' = 3Ae^{3x}$, $y_p'' = 9Ae^{3x}$. Substituting: $9A + 3A - 2A = 6 \Rightarrow 10A = 6 \Rightarrow A = \frac{3}{5}$. Thus $y_p = \frac{3}{5}e^{3x}$.

6 Find a particular solution of $y'' - 4y = \cos x$.

Try $y_p = A\cos x + B\sin x$: $y_p'' = -A\cos x - B\sin x$. Substituting: $-A\cos x - B\sin x - 4A\cos x - 4B\sin x = \cos x \Rightarrow -5A\cos x - 5B\sin x = \cos x$. So $A = -\frac{1}{5}$, $B = 0$. Thus $y_p = -\frac{1}{5}\cos x$.

7 Find a particular solution of $y'' - y' - 2y = x^2$.

Try $y_p = Ax^2 + Bx + C$: $y_p' = 2Ax + B$, $y_p'' = 2A$. Substituting: $2A - (2Ax + B) - 2(Ax^2 + Bx + C) = x^2 \Rightarrow -2Ax^2 + (-2A - 2B)x + (2A - B - 2C) = x^2$. Matching coefficients: $-2A = 1 \Rightarrow A = -\frac{1}{2}$; $-2A - 2B = 0 \Rightarrow B = -A = \frac{1}{2}$; $2A - B - 2C = 0 \Rightarrow -1 - \frac{1}{2} - 2C = 0 \Rightarrow C = -\frac{3}{4}$. Thus $y_p = -\frac{1}{2}x^2 + \frac{1}{2}x - \frac{3}{4}$.

The modification rule (resonance)

8 Find a particular solution of $y'' - 3y' + 2y = e^{2x}$. (Note $r = 2$ IS a root of $r^2 - 3r + 2$.)

Since e^{2x} solves the homogeneous equation, try $y_p = Axe^{2x}$. Then $y_p' = Ae^{2x} + 2Axe^{2x}$ and $y_p'' = 4Ae^{2x} + 4Axe^{2x}$. Substituting: $(4Ae^{2x} + 4Axe^{2x}) - 3(Ae^{2x} + 2Axe^{2x}) + 2Axe^{2x} = Ae^{2x}$ (the xe^{2x} terms cancel). Setting $Ae^{2x} = e^{2x}$ gives $A = 1$. Thus $y_p = xe^{2x}$.

9 Find a particular solution of $y'' - 4y' + 4y = e^{2x}$. (Note $r = 2$ is a REPEATED root, so the usual modification xe^{2x} isn't enough.)

Since both e^{2x} and xe^{2x} solve the homogeneous equation (repeated root $r = 2$), try $y_p = Ax^2e^{2x}$. Computing $y_p' = 2Axe^{2x} + 2Ax^2e^{2x}$ and $y_p'' = 2Ae^{2x} + 8Axe^{2x} + 4Ax^2e^{2x}$, substitution gives $y_p'' - 4y_p' + 4y_p = 2Ae^{2x}$ (all xe^{2x} and x^2e^{2x} terms cancel). Setting $2Ae^{2x} = e^{2x}$: $A = \frac{1}{2}$. Thus $y_p = \frac{1}{2}x^2e^{2x}$.

Combining forcing terms

10 Find a particular solution of $y'' - y = e^x + x$. (Roots of $r^2 - 1 = 0$ are $r = \pm 1$, so e^x resonates but x does not.)

For the e^x part (resonant): try Axe^x, giving $y_{p1}'' - y_{p1} = 2Ae^x = e^x \Rightarrow A = \frac{1}{2}$, so $y_{p1} = \frac{1}{2}xe^x$. For the x part: try $Bx + C$, giving $-(Bx + C) = x \Rightarrow B = -1, C = 0$, so $y_{p2} = -x$. Total: $y_p = \frac{1}{2}xe^x - x$.

11 Find a particular solution of $y'' + y = 3\sin x + 2$. (The forcing has both a resonant sine term and a non-resonant constant.)

For the 2: try C, giving $C = 2$. For $3\sin x$ (since $\sin x, \cos x$ solve $y'' + y = 0$): try $x(A\cos x + B\sin x)$. Computing $y_p'' + y_p$ for this piece leaves $-2A\sin x + 2B\cos x$ (the $x\sin x, x\cos x$ terms cancel). Setting $-2A\sin x + 2B\cos x = 3\sin x$: $A = -\frac{3}{2}$, $B = 0$. Total: $y_p = -\frac{3}{2}x\cos x + 2$.

Solving full initial value problems

12 Solve $y'' - y = 4x$, $y(0) = 1$, $y'(0) = 1$.

$y_c = C_1e^x + C_2e^{-x}$. Try $y_p = Ax$: $-Ax = 4x \Rightarrow A = -4$, so $y_p = -4x$. General: $y = C_1e^x + C_2e^{-x} - 4x$. $y(0) = C_1 + C_2 = 1$. $y' = C_1e^x - C_2e^{-x} - 4$; $y'(0) = C_1 - C_2 - 4 = 1 \Rightarrow C_1 - C_2 = 5$. Adding to $C_1 + C_2 = 1$: $2C_1 = 6 \Rightarrow C_1 = 3$, $C_2 = -2$. Thus $y = 3e^x - 2e^{-x} - 4x$.

13 Solve $y'' + y = 2\cos x$, $y(0) = 0$, $y'(0) = 0$. (This is a resonant forcing.)

$y_c = C_1\cos x + C_2\sin x$. Since $\cos x$ resonates, try $y_p = x(A\cos x + B\sin x)$; substitution gives $y_p'' + y_p = -2A\sin x + 2B\cos x$. Setting this $= 2\cos x$: $A = 0$, $B = 1$, so $y_p = x\sin x$. General: $y = C_1\cos x + C_2\sin x + x\sin x$. $y(0) = C_1 = 0$. $y' = -C_1\sin x + C_2\cos x + \sin x + x\cos x$; $y'(0) = C_2 = 0$. Thus $y = x\sin x$ — the amplitude grows without bound, the signature of resonance.

Application: a driven oscillator

14 A forced oscillator satisfies $y'' + 4y = \sin 3t$ (forcing frequency $3 \neq$ natural frequency 2 — no resonance). Find a particular solution.

Try $y_p = A\sin 3t + B\cos 3t$: $y_p'' = -9A\sin 3t - 9B\cos 3t$. Substituting: $-5A\sin 3t - 5B\cos 3t = \sin 3t \Rightarrow A = -\frac{1}{5}$, $B = 0$. Thus $y_p = -\frac{1}{5}\sin 3t$.

15 Now solve $y'' + 4y = \sin 2t$, where the forcing frequency EQUALS the natural frequency (resonance). Find a particular solution.

Since $\sin 2t, \cos 2t$ solve the homogeneous equation, try $y_p = t(A\sin 2t + B\cos 2t)$. Computing y_p'' and substituting, the terms proportional to $t\sin 2t$ and $t\cos 2t$ cancel, leaving $y_p'' + 4y_p = 4A\cos 2t - 4B\sin 2t$. Setting this equal to $\sin 2t$ gives $A = 0$, $B = -\frac{1}{4}$. Thus $y_p = -\frac{1}{4}t\cos 2t$ — the amplitude grows linearly in t, the signature of resonance.

16 For the non-resonant case $y'' + 4y = \sin 3t$, use the particular solution above to state the full general solution.

$y_c = C_1\cos 2t + C_2\sin 2t$, so $y = C_1\cos 2t + C_2\sin 2t - \frac{1}{5}\sin 3t$.

Choosing the correct trial form

17 State the correct (unmodified) trial form for a particular solution of $y'' - y' - 6y = 5e^{4x}$, and explain why no resonance check is needed here.

Try $y_p = Ae^{4x}$. Since the auxiliary equation $r^2 - r - 6 = (r-3)(r+2) = 0$ has roots $r = 3, -2$, and 4 is neither, e^{4x} does not solve the homogeneous equation, so the unmodified exponential trial form applies directly.

18 State the correct trial form for a particular solution of $y'' - y' - 6y = 5e^{3x}$, explaining why it must be modified.

Since $r = 3$ IS a root of $r^2 - r - 6 = 0$, the unmodified trial Ae^{3x} would solve the homogeneous equation and could never balance a nonzero right side. The correct trial form is $y_p = Axe^{3x}$.

19 State the correct trial form for a particular solution of $y'' + 9y = x\sin 3x$. (Note $\sin 3x$ resonates with the homogeneous solutions.)

Since $\cos 3x, \sin 3x$ solve $y'' + 9y = 0$, and the forcing is $x\sin 3x$ (a degree-1 polynomial times a resonant trig function), the trial form must combine BOTH modifications: $y_p = x(Ax+B)\cos 3x + x(Cx+D)\sin 3x$, i.e., $y_p = (Ax^2+Bx)\cos 3x + (Cx^2+Dx)\sin 3x$.

20 State the correct trial form for a particular solution of $y'' - 2y' + y = 3xe^x$. (Note $r = 1$ is a repeated root of the homogeneous equation.)

Since e^x and xe^x both solve the homogeneous equation (repeated root $r = 1$), and the forcing $3xe^x$ is degree 1 times e^x, the trial form needs an extra factor of x^2: $y_p = x^2(Ax+B)e^x = (Ax^3+Bx^2)e^x$.

The method's limits

21 Explain why undetermined coefficients cannot be used directly on $y'' + y = \tan x$.

The method only works when the forcing term belongs to one of a fixed list of families (polynomials, exponentials, sines/cosines, and products of these) whose derivatives stay within a finite-dimensional span — so a trial form with finitely many unknown coefficients can absorb every derivative that substitution produces. $\tan x$ is not in this list: its derivatives generate an ever-growing family of new functions ($\sec^2 x$, $\sec^2 x \tan x$, ...), so no finite trial form exists. Variation of parameters is needed instead.

22 Explain why undetermined coefficients cannot be used directly on $y'' + y = \frac{1}{x}$, even though $\frac{1}{x}$ looks like a simple algebraic expression.

The allowed forcing families for undetermined coefficients are polynomials, exponentials, sines/cosines, and their products — not general rational functions. $\frac{1}{x} = x^{-1}$ is not a polynomial (negative exponent), so its repeated differentiation doesn't stay inside a finite-dimensional trial family. Variation of parameters handles this case instead.

23 For $y'' - y = xe^x \ln x$ ($x > 0$), explain briefly why undetermined coefficients fails but variation of parameters still applies.

The factor $\ln x$ is outside the finite list of forcing types (polynomial/exponential/sine-cosine and their products) that undetermined coefficients can match with a finite trial form — differentiating $\ln x$ never produces a polynomial multiple of itself. Variation of parameters places no restriction on the forcing term's form at all (only on being able to evaluate the resulting integrals), so it still applies.

24 Is $y'' + y = \sec x$ (for $-\frac{\pi}{2} < x < \frac{\pi}{2}$) solvable by undetermined coefficients? Justify your answer.

No. $\sec x = \frac{1}{\cos x}$ is not a polynomial, exponential, sine/cosine, or product of these, so repeated differentiation does not stay within a finite trial family (it generates $\sec x \tan x$, $\sec^3 x$, and further terms). Variation of parameters is required.

25 Is $y'' - 4y = x^2 e^{3x} \cos x$ solvable by undetermined coefficients? If so, describe (without solving) the shape of the trial form.

Yes — it is a product of a polynomial (x^2), an exponential (e^{3x}), and a trig function ($\cos x$), which is exactly the family undetermined coefficients handles. Since $r = 3 \pm i$ (loosely speaking, the exponential-times-trig factor $e^{3x} \cos x$) is not itself a root of $r^2 - 4 = 0$, no resonance modification is needed, and the trial form is $y_p = (Ax^2 + Bx + C)e^{3x} \cos x + (Dx^2 + Ex + F)e^{3x} \sin x$.